Home Automation

Project Proposal for the course
CSE 342: IoT based Project Development

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Abstract

Home Automation
Home automation is the use of technology to improve the quality and comfort of human living standards. Out of thousands of home automation systems available, this project considers controlling lights and fans. This project gives users the power to control these devices all over the world using the Internet; it is not only convenient but also energy-saving. An important characteristic of this system is the ability to control fan speed smoothly. The app Blynk has a very easy-to-use graphical interface for user interaction. Apart from this, users have the option to control these appliances manually. Overall, this project aims at improving the level of comfort within the house, helping users with their daily chores, and increasing the convenience factor for the user.

Contents

Abstract **1**

1 Introduction **3**
1.1 Problem Statement 3
1.2 Scope 3
1.3 Outcomes/Objectives 3

2 Methodology or Analysis and Design **4**
2.1 Methodology 4
2.2 Design 4
2.2.1 Problem Statement 4
2.2.2 Requirement Analysis 4
2.2.3 Use cases 5
2.2.4 Design Diagrams 5

3 Implementation **8**
3.1 Requirements 8
3.2 Method 8
3.3 Evaluation Measures 9

4 Conclusions **10**
4.1 Achievements 10
4.2 Recommendations 11

Chapter 1

Introduction

1.1 Problem Statement

The traditional home consists of manual operation of electronic devices that lead to energy waste and inconvenience. People frequently forget to turn off the lights and fans, which results in excess energy consumption and higher electric bills. Besides, manually operating switches might be difficult for those with physical disabilities or elderly people.

1.2 Scope

This setup is designed particularly for small-scale applications where a stable internet connection is required. It can be operated manually as well as remotely.

1.3 Outcomes/Objectives

The objective of this project is to design and build a smart home automation system that will enable users to manage 220V AC lighting and a 12V DC fan from a distance. This system aims to make things more convenient for everyone while consuming less energy.

Chapter 2

Methodology or Analysis and Design

2.1 Methodology

The initial phase entails the selection of components according to the requirements. Next, the integration of the system is performed, which involves connection of the system components and configuration of the PWM signals to control the speed of the fan. After the hardware is configured, the next step is the software development part, where coding of the ESP32 microcontroller is done through the Arduino IDE and the Blynk app. Finally, confirm system communication, as well as response time.

2.2 Design

2.2.1 Problem Statement

Since this system completely depends on IoT, Wi-Fi connection issues or weak signals may lead to delays in command execution.

2.2.2 Requirement Analysis

Hardware Components:

- ESP32 microcontroller
- 2-channel relay module
- L298N DC motor driver

- 12V DC fan
- 220V AC bulbs
- Power supply adapter
- Push buttons for manual control
- Register

Software Components:

- Blynk App for IoT control
- Arduino IDE for ESP32 programming

2.2.3 Use cases

- **The Esp32 microcontroller** is used to process user commands from the Blynk app or physical buttons and also sends signals to the relay module and the the L298N motor driver to control the light and fan.
- **2-channel relay module** works as an electronic switch, which allows esp32 to turn on/off lights.
- **L298N DC Motor Driver** controls the speed of the fan by regulating the voltage supplied to the DC motor using PWM signals.
- **Blynk App** provides an interface for remote control.

2.2.4 Design Diagrams

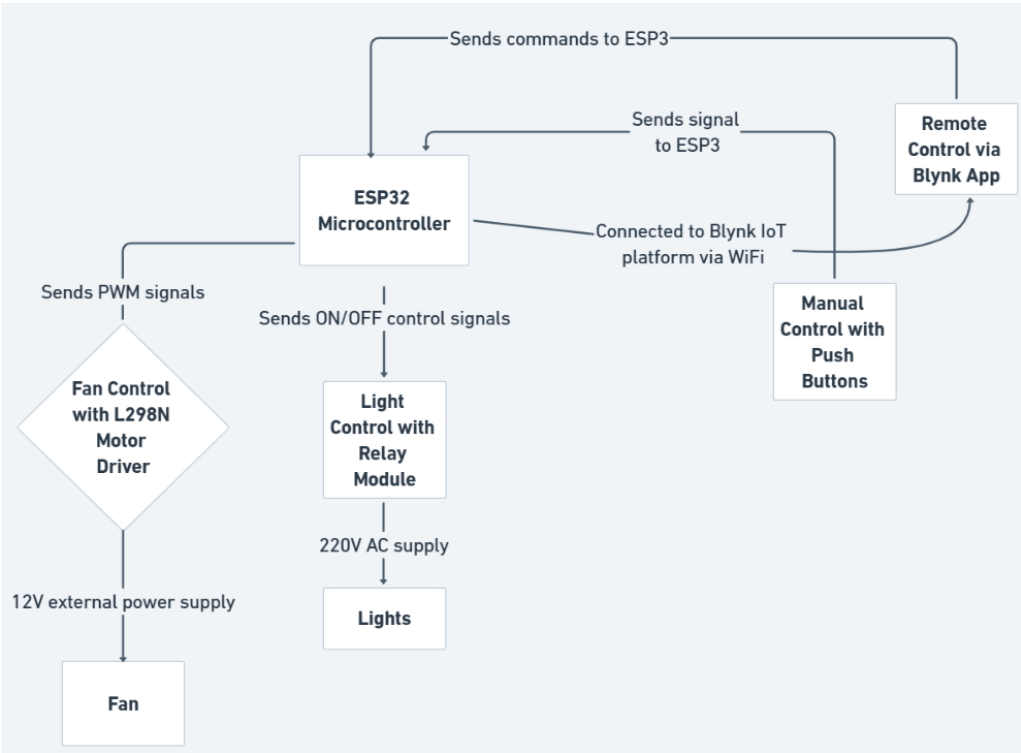

Figure 2.1: ESP32-Based Smart Home Control Flowchart

This flowchart shows a smart fan and light control system using an ESP32 microcontroller. Connects to the Blynk IoT platform via WiFi for remote access and also allows manual control via push buttons. The ESP32 transmits PWM signals to an L298N motor driver, which regulates the speed of a fan powered by a 12V external source. It also sends ON/OFF control signals to a relay module, which switches lights operated by a 220V AC supply. Users can control the system remotely using the Blynk app or manually using push buttons.

Chapter 3

Implementation

3.1 Requirements

You need to program ESP32 using Arduino IDE, Blynk App setup with user interface controls, and configuration of hardware components.

3.2 Method

To connect the system, power the ESP32 via USB, the L298N motor driver with a 12V DC source, and the relay module with the 5V pin of the ESP32.

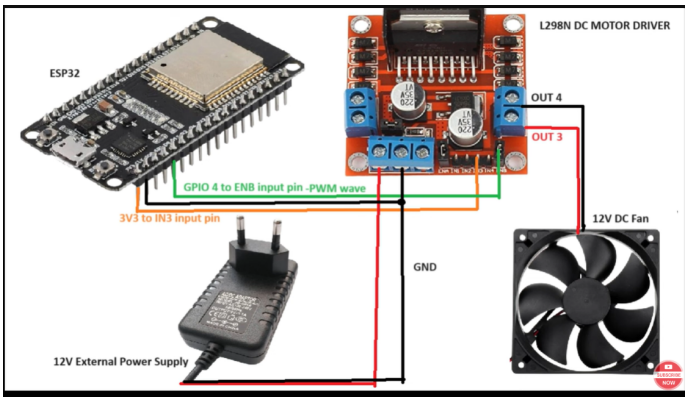

Figure 3.1: Connection diagram for fan control [1]

For fan control, connect ESP32 GPIO 4 to ENB pin of L298N for PWM speed control, ESP32 3V3 to IN3 for enabling it, and OUT3 and OUT4 of the motor driver to the fan. [1]

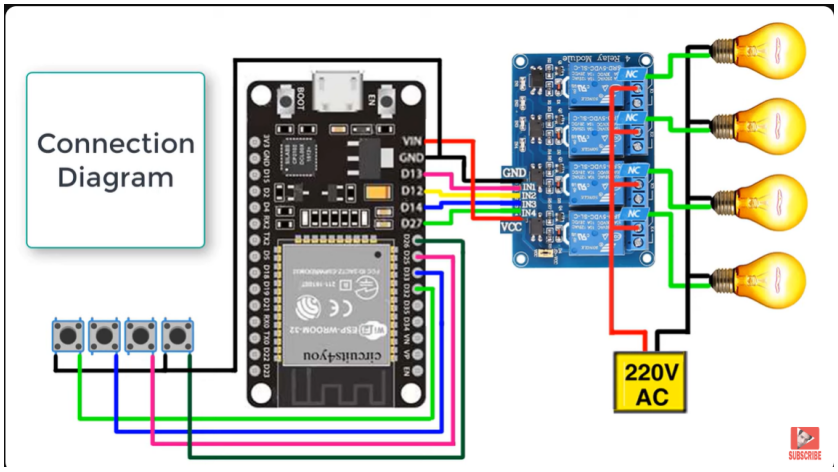

Figure 3.2: Connection diagram for light control [2]

This circuit is for **light control**,supplied from a 220V AC supply, which is most likely reduced to a lower DC voltage to supply power through VIN and VCC pins, with GND as common ground. Digital pins D12, D14, and D22 on a microcontroller are used to interface with external devices, enabling control or data exchange.[2]

The ESP32 connects to WiFi and the Blynk app, where virtual pins send ON/OFF signals for the relay and PWM values for fan speed, enabling manual and remote control.

3.3 Evaluation Measures

Measurement of performance is to evaluate the ability of users to control lights or fans via the Blynk command application. The fan speed controller features slider controls for a more precisely controlled level of speed selection, which in turn will show numerical representation of current speed in real-time. And similarly, the light control buttons on the display offer a choice between utilizing virtual ON or OFF switches to turn lights ON and OFF.

Chapter 4

Conclusions

To conclude, there are many important advantages to using smart technology at home in a home automation project. We can automate home tasks such as controlling the lighting and temperature of our rooms, keeping an eye on security, and more from anywhere in the world. During the project, we were able to gain knowledge about real-time data processing, secure communication protocols, and system integration. This project simplifies daily chores and reduces energy use. It also allows the homeowner to manage their living spaces. We also understood how important user-friendly interfaces are in facilitating smooth communication with automated processes. This study not only improved our understanding of IoT frameworks, but also demonstrated the practical applications of these technologies.

4.1 Achievements

Effective use of IoT technologies to deploy a smart home automation system. Automatic control of important home operations, including controlling lights, adjusting indoor temperatures, and maintaining security systems without human manual intervention. Performs immediate actions and a range of sensors to identify environmental factors, such as turning off lights in a room when no one is using it. A user-friendly interface for remote control and system monitoring is being developed which provides easy access to system settings and status update. A better understanding of IoT protocols, cloud connection, and data security measures which need to protect private information from cyberattacks.

4.2 Recommendations

Protect home automation system by implementing strong passwords and secure network. Make familiar with the use of operating system to household members, so that they can access the system.Select home appliances that lower utility bills and energy consumption. Select appliances that supports future upgrades. Use a platform to control all smart appliances through smartphone.Abstract Home automation is the use of technology to improve the quality and comfort of human living standards. Out of thousands of home automation systems available, this project considers controlling lights and fans. This project gives users the power to control these devices all over the world using the Internet; it is not only convenient but also energy-saving. An important characteristic of this system is the ability to control fan speed smoothly. The app Blynk has a very easy-to-use graphical interface for user interaction. Apart from this, users have the option to control these appliances manually. Overall, this project aims at improving the level of comfort within the house, helping users with their daily chores, and increasing the convenience factor for the user.

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[2] Ahmad Logs. Home automation system using esp32 and blynk 2.0 — blynk esp32 relay control. YouTube Video, 2021. Accessed: 2025-03-14.